Eggshell–Agar Biocomposites

Development and characterisation for sustainable interior applications

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The waste



94%
calcium carbonate

> 12
months of collection

1
local bakery

Porous, protein-bound crystalline structure. Specific gravity 0.85–2.66 — lighter than most mineral fillers.

The problem

ESTABLISHED PRACTICE

Mineral phase

waste eggshell, 5–30% of matrix

Binding phase

Portland clinker · bitumen · lime · ceramic body · PLA, epoxy, PVC, PU

The waste is valorised. The matrix is not.

THIS STUDY

Mineral phase

waste eggshell, 66% of solids

Binding phase

agar, plasticised with glycerol

Both phases biological and biodegradable.

Teara et al. 2019 · Adogla et al. 2016 · Razzaq & Yousif 2017 · Soundara & Vilasini 2015 · Vilarinho et al. 2022 · Owuamanam & Cree 2020

The binder

Agar is a thermoreversible polysaccharide from red seaweed. Its behaviour with glycerol has been mapped systematically — 32 formulations, released as an open dataset.

Hernández et al. (2022), Materials 15(11), 3954

BUT ONLY AS A FILM

Unfilled, thin, tested in uniaxial tension, for packaging. Never as the binder of a mineral-filled cast solid in bending.

THE SHIFT

—	Film literature	This study
Form	Thin cast film	Cast solid, 20–50 mm
Filler	Unfilled	Mineral, 66% of solids
Loading	Uniaxial tension	Compression, bending
Use	Packaging	Interior surfaces

The compositional logic carries over. The form and the loading case do not.

The gap

Materiom publishes biomaterial recipes under CC-BY-SA, archives each as prior art, and indexes property data with citable DOIs. The infrastructure for shared procedures is built.

4

recipes with published property data

WHICH ARE

agar bioplastic
gelatin bioplastic
mycelium–walnut shell
mussel shell–sucrose ceramic

WHAT THAT MEANS

The nearest analogue is the mussel shell–sucrose ceramic: a shell mineral in a biological binder. Nothing couples a shell mineral to agar. Nothing reports compressive, flexural, ignitability and hygric data for one formulation family.

A protocol and a property set entering that commons.

materiom.org/data, accessed October 2026

Questions

01 Which formulation variable governs compressive strength?

02 What does plasticiser content do to flexural behaviour?

03 How does it compare with conventional interior materials?

04 How does it compare with other bio-based composites?

05 Is compressive strength the right criterion at all?



Four constituents, all biodegradable

Waste eggshell powder · agar · glycerol · water, with a shellac finish. Colour and texture vary from natural pigments and mould geometry at constant composition.

What each does

Constituent	Origin	Function	Relevant property
Eggshell powder	Food waste	Mineral filler; carries load	94% $CaCO_3$; porous structure
Agar	Red seaweed	Continuous binding phase	Thermoreversible gel
Glycerol	Vegetable oil	Plasticiser; sets ductility	Humectant; low vapour pressure
Water	—	Carrier; sets solids packing	Fully evaporated on drying
Shellac	Kerria lacca	Hydrophobic surface finish	Biodegradable film-former

Water is not in the finished material. It is the process variable that sets how densely the solids pack.

Gelatine, starch and vinegar were trialled during development and are not part of the reported formulations.

Protocol

120 °C	95 °C	80 °C	23 °C
20 min		55 min total	11 days
01 **Mineral fraction**	02 **Binder phase**	03 **Compositing**	04 **Casting and cure**
– Clean of membrane – Heat-treat, then dry – Mill to fine powder – Sieve, 150 µm passing	– Water and glycerol first – Agar added incrementally – Held until fully dissolved	– Temperature reduced – Viscosity develops – Shell added incrementally	– Silicone moulds – Four days in mould – Seven days ambient – No oven at any stage

Disclosure — the composite shrinks appreciably as water evaporates; post-drying dimensions were not logged, so dimensions quoted are mould dimensions. Stirring was manual: shear rate is not a controlled quantity.

Formulations

ID	Test	Water g	Shell g	Agar g	Glycerol g	Solids g	Shell : binder	Solids : water	Glycerol % solids
C1	Compression	325	80	35	5	120	**2.00**	0.37	4.2
C2	Compression	310	90	40	5	135	**2.00**	0.44	3.7
C3	Compression	280	110	45	10	165	**2.00**	0.59	6.1
F1	Flexure	450	190	70	60	320	1.46	0.71	18.8
F2	Flexure	480	190	70	30	290	1.90	0.60	10.3

COMPRESSION SERIES

Shell : binder is held at exactly 2.00. The only variable is solids concentration.

FLEXURAL SERIES — CONFOUND

Glycerol was halved at fixed batch mass, so water rose with it. The plasticiser effect is not isolated.

Three specimens per formulation. Values are means; individual specimen values were not retained, so standard deviations are not available.

Tests

Property	Standard	Specimen	n	Measured
Compressive behaviour	Cube compression	50 mm cube moulds	3 per mix	Max load; compressive stress
Flexural behaviour	ASTM D790-17	180 × 60 × 20 mm, 120 mm span	3 per mix	Flexural strength; modulus
Reaction to fire	ISO 11925-2:2020	40 × 50 mm, 45°, 15 s edge flame	2	Time to ignition; to extinguish
Water absorption	ASTM D570-98	40 × 50 mm, shellac-coated	1	Mass gain, 24 h immersion
Bulk density	Gravimetric	50 mm cube	1	Mass / mould volume

Mechanical values are means of three specimens; individual values were not retained, so standard deviations cannot be reported. Fire, absorption and density rest on single determinations and are indicative, not characteristic.

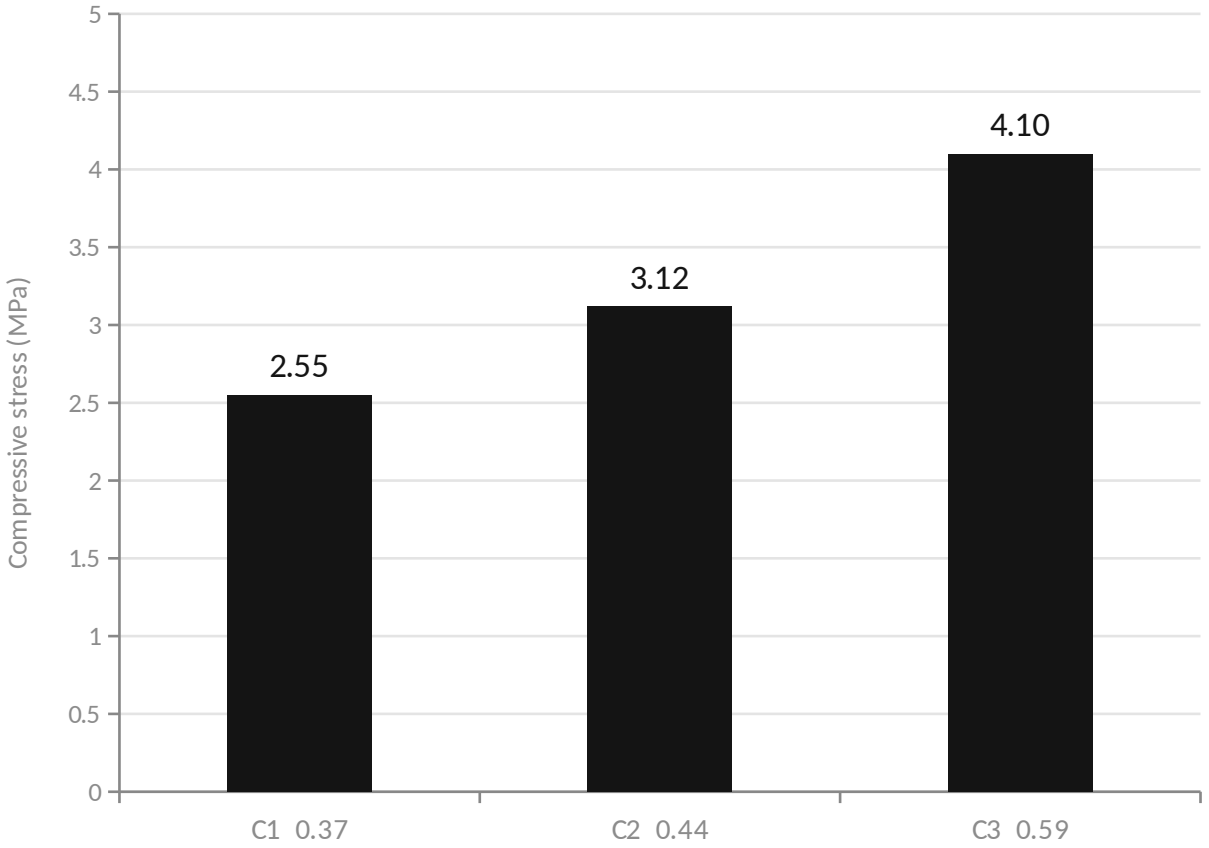
All testing: Advanced Construction Materials Research Laboratory, Hacettepe University.

Compression

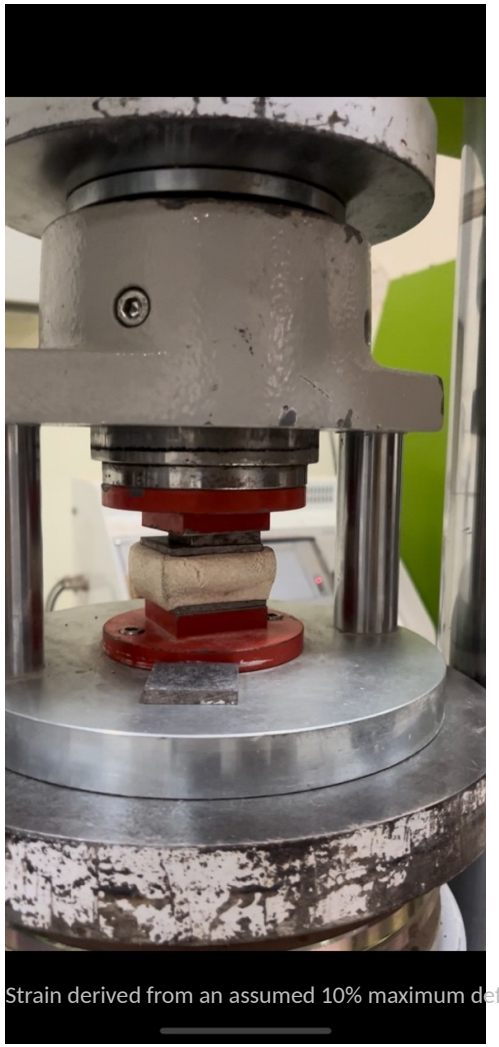
ID	Solids : water	F_{max} kN	σc MPa
C1	0.37	3.89	2.55
C2	0.44	4.30	3.12
C3	0.59	**6.80**	**4.10**

Strength rises with solids concentration, not with mineral content.

Shell : binder is fixed at 2.00 across the series, so packing density is the only variable. Linear across the range tested, $R^2 = 0.99$ — three formulations, so a hypothesis, not a law.

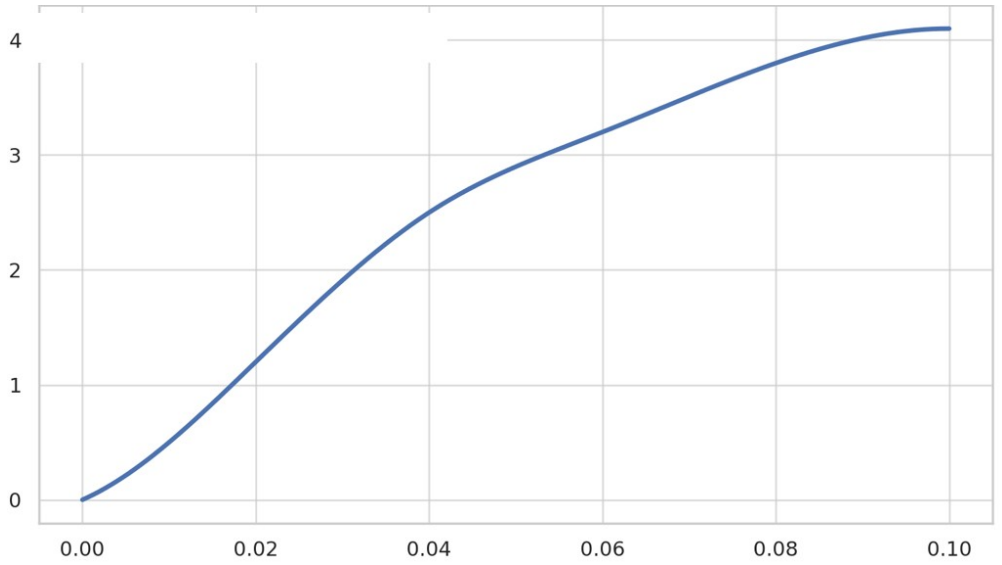


Crushing



Strain derived from an assumed 10% maximum deformation based on observation, not instrumented measurement. · Bulk density 1.38 g/cm³, mould-volume basis.

COMPRESSIVE STRESS – STRAIN, C3



THE CURVE

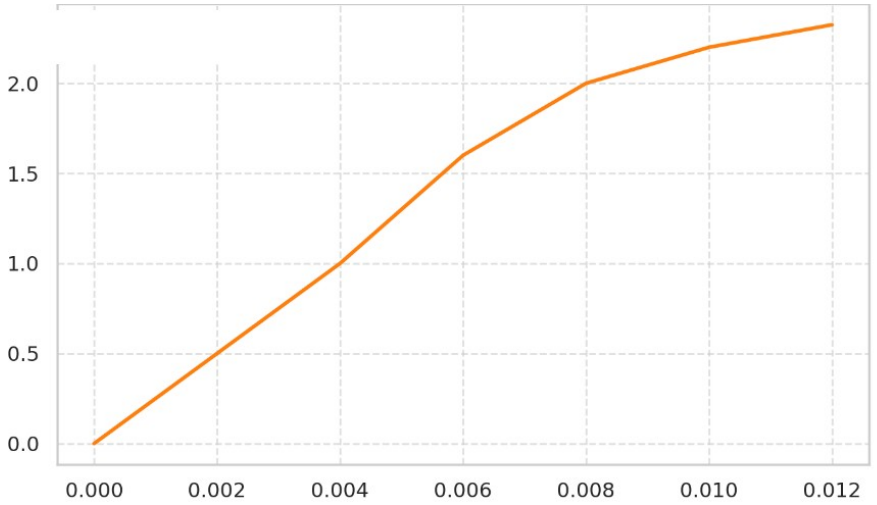
- Linear to about 1 MPa
- Progressive stiffness loss
- Peak near 10% strain
- No abrupt load drop
- Deformation with retained integrity

It crushes.
It does not fracture.

Flexure

ID	Glycerol % solids	F_{max} N	σF MPa	EF MPa	ε peak
F1	18.8	—	*not measurable*	—	—
F2	10.3	310	**2.325**	**192**	0.0121

FLEXURAL STRESS – STRAIN, F2



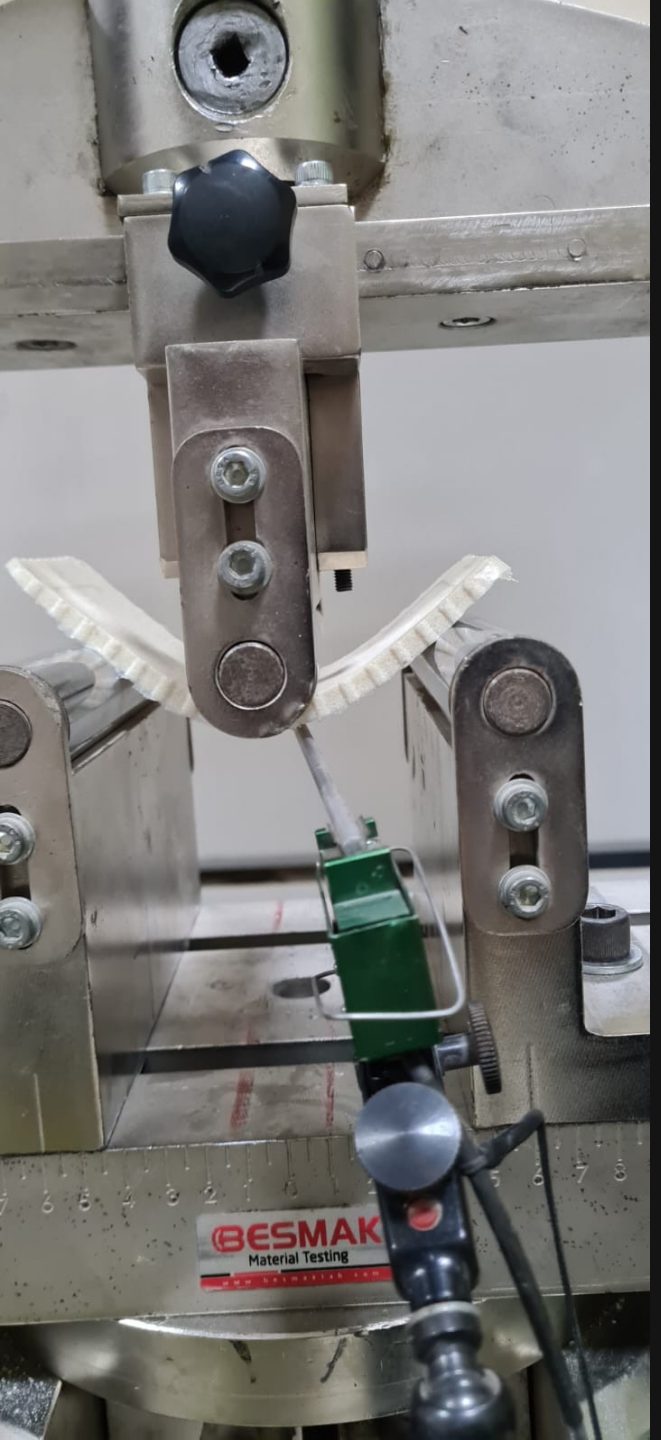
Three specimens per formulation. The two formulations differ in glycerol, but batch mass was fixed, so solids concentration moved as well.

DISCLOSURE

The specimen is kerfed on its tension face. The standard assumes a solid rectangle, and the calculation uses the full section. Part of the compliance is geometric: the modulus is a lower bound.

Plasticiser decides whether it resists at all.

At 18.8% glycerol the machine registered nothing usable. Halving it gave a section that resists while still flexing.







BENDING

The kerfs are visible, and so is the consequence.

Transverse kerfs on the tension face let the panel arch by hand. They also mean the measured compliance is partly geometric, which is why the reported modulus is a lower bound rather than a material constant.

Three-point rig at high deflection · close-up of the kerfed tension face · panel arched by hand

Fire



~10 s

to ignition under direct flame

4 s

to self-extinguish after the flame is removed

- Charring stays local
- No lateral spread
- No melting, no dripping

MECHANISM

The carbonate fraction decomposes endothermically and releases CO_2, diluting the flame front locally. Consistent with the absence of spread, but not verified calorimetrically.

WHAT THIS IS NOT

ISO 11925-2 is an ignitability test. No Euroclass classification was determined. No cone calorimetry, smoke density or toxicity data were obtained.

Water and weight

ABSORPTION — 24 H FULL IMMERSION

0.7%

mass gain, shellac-coated

Local dissolution where the coating was thin. Coating integrity, not the composite, is the limiting factor for humid use.

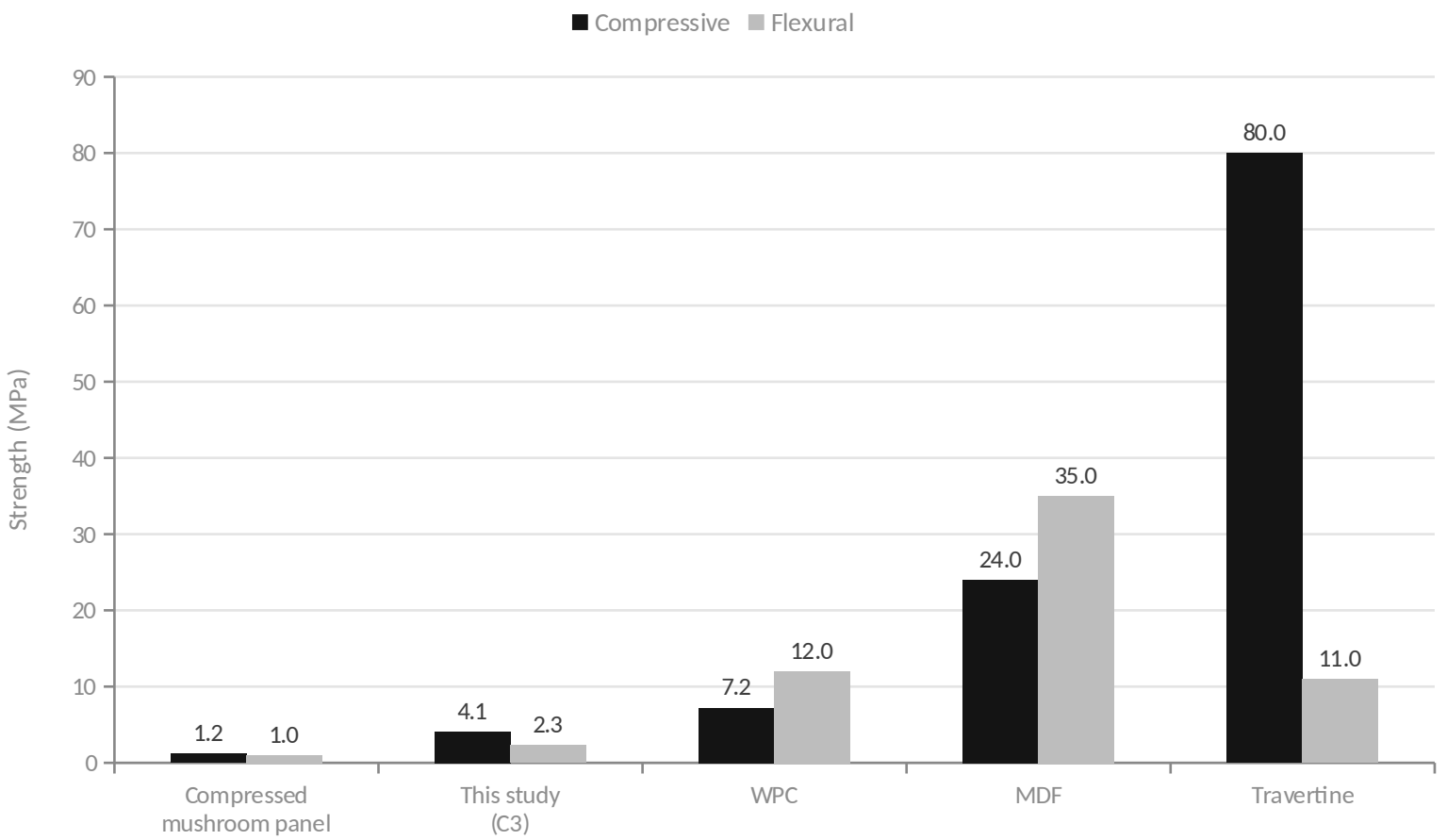
Three coats of shellac — a natural, biodegradable, hydrophobic resin. Slight dimensional expansion was observed.

BULK DENSITY

Material	g/cm³	Reference
This study (C3)	**1.38**	—
Mycelium, cellulose	0.3 – 0.8	Jones et al. 2020
Fired clay brick	1.6 – 2.0	William & Owen 1995
Normal-weight concrete	2.4	William & Owen 1995

Denser than the fungal and cellulosic family, lighter than fired clay and concrete. Quoted on a mould-volume basis and therefore a lower bound.

Versus stone and board



Velasco & Gaona 2011 · Klyosov 2007 · Nemli & Aydın 2007 · Altun 2006

On strength alone it does not compete.

One sixth the compressive strength of MDF. One twentieth that of travertine — a stone it visually resembles. Three times compressed mushroom panel, the only other fully bio-based board here.

Versus bio-based

COMPRESSIVE STRENGTH

Material	σc MPa	Reference
This study (C3)	**4.1**	—
Mycelium composite	0.145 – 0.452	Jones et al. 2020
Hempcrete	0.3 – 0.6	Arnaud & Gourlay 2012
Eggshell-filled PLA	2 – 5	Kim & Pal 2010
Mineral-filled biocomposite	2.1 – 39.6	Shah et al. 2016

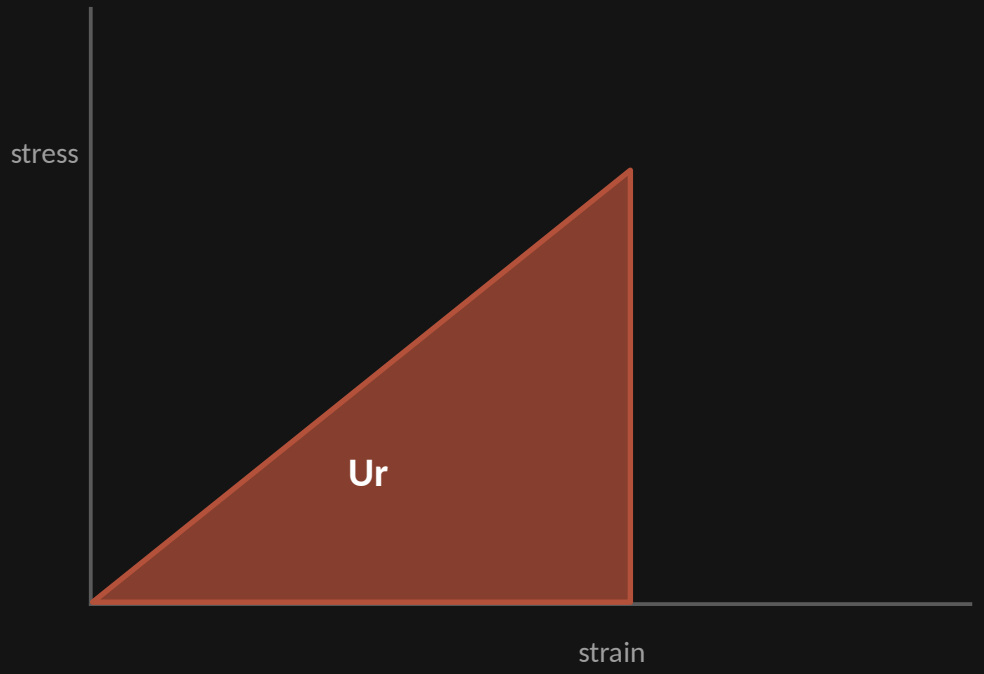
ELASTIC MODULUS

Material	E MPa	Reference
This study (F2)	**192**	—
Mycelium composite	1.8 – 5	Islam et al. 2018
Soy-based bioplastic	124	Mansour et al. 2021
Flax-fibre PLA	198	Faruk et al. 2012
LDPE	100 – 300	Zhao et al. 2022
Wood (pine)	11 000	Pope & Hudson 1989

Water-cast and ambient-dried, at melt-processed stiffness.

An order of magnitude above the fungal and lime-binder family in compression, and inside the bio-polymer band in stiffness — between soy bioplastic and flax-fibre PLA, within the LDPE range. No oven, no autoclave, no extrusion. Wood remains two orders of magnitude above.

Resilience



The shaded area is the elastic energy absorbed per unit volume before permanent damage.

$$Ur = \sigma^2 / 2E$$

14.1 kJ/m³

for the formulation measured here

Stiffness sits in the denominator.

The low modulus that counted against the material on the previous slide is exactly what raises this quantity.

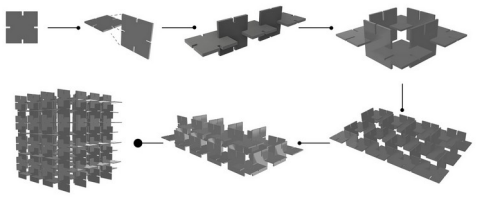
THE LIMIT

A comparative table cannot yet be assembled: the literature reports strength and modulus in separate studies on different formulations. This is a data point and a measurement convention, not a ranking.

Where it fits



Modular interior proposal developed from the measured properties



Assembled without adhesive, so components can be recovered intact. Units interlock and share load. A damaged unit is exchanged without dismantling.

Non-loadbearing surfaces

self-weight and handling loads only

Dry interiors

not wet rooms, not facades

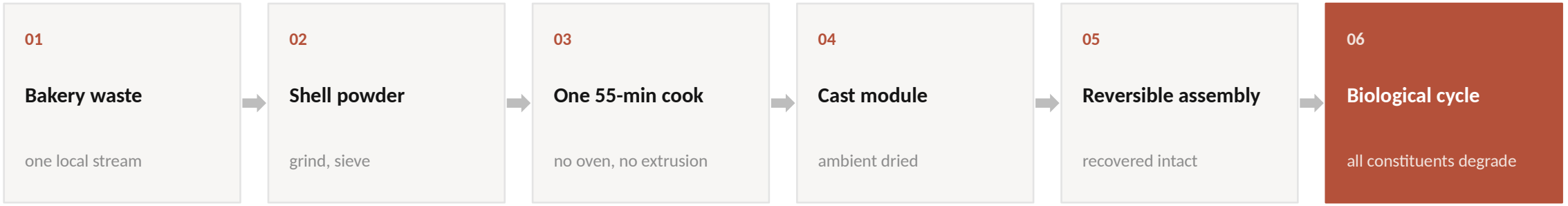
Away from ignition sources

self-extinguishing, but unclassified

Modular, not sheet

failure replaced one unit at a time

Why it matters



The loop closes: the material returns to the stream it came from.

WHAT IS DISPLACED

Quarried limestone, in a surface application rather than a downcycled one. A grinder, a sieve, a hotplate and silicone moulds: production can be distributed rather than centralised.

No life-cycle assessment was carried out. These are qualitative potentials grounded in the protocol, not quantified environmental claims.

Limits

No dispersion data

Means of three specimens; individual values not retained.

Confounded flexural series

Glycerol, solids concentration and shell : binder moved together.

Kerfed section, solid assumption

The modulus is a lower bound, not a material constant.

Strain not instrumented

Compressive strain assumed from observation.

Shrinkage unquantified

Density is a lower bound on a mould-volume basis.

Process not instrumented

Manual stirring; one laboratory, one operator.

No durability testing

No humidity or thermal cycling, UV, microbial or ageing.

No fire class, no LCA

Ignitability only; no quantified environmental assessment.

Each defines a specific next experiment.

Conclusions

01

A protocol others can repeat

Every parameter reported: 120 °C · 150 µm · 95 °C · 80 °C · 55 min · silicone · 11 days ambient

02

Packing density governs strength

With shell : binder fixed, compressive stress rises linearly with solids concentration

03

Plasticiser sets brittle or ductile

Between 18.8% and 10.3% glycerol, from unmeasurable to 2.33 MPa at 192 MPa

04

Bio-polymer stiffness, oven-free

Water-cast and ambient-dried, inside the melt-processed bio-polymer band

05

Strength is not the only criterion

Deformation capacity governs service behaviour; report σ and E together

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Testing: Advanced Construction Materials Research Laboratory, Hacettepe University, with thanks to Dr Hüseyin İlcan. Derived from the author's doctoral dissertation, Hacettepe University, 2025.

References

REVIEWS AND STATE OF THE ART

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- Owuamanam, S. & Cree, D. (2020). Progress of bio-calcium carbonate waste eggshell and seashell fillers in polymer composites. Journal of Composites Science, 4(2), 70.
- Biogenic calcium carbonate from waste shells for advanced material applications (2022). Frontiers in Materials, 9, 1024977.

AGAR AND BIO-POLYMER SYSTEMS

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- Materiom (2026). Open biomaterial recipe commons and property database. materiom.org/data.
- Mansour et al. (2021) · Faruk et al. (2012) · Zhao et al. (2022) — bio-polymer elastic moduli.

COMPARATIVE MATERIAL DATA

- Jones, M. et al. (2020) · Islam et al. (2018) — mycelium composites. Arnaud & Gourlay (2012) — hempcrete.
- Kim & Pal (2010) — eggshell-filled PLA. Shah et al. (2016) — mineral-filled biocomposites.
- Nemli & Aydın (2007) — MDF. Altun (2006) — travertine. Klyosov (2007) — WPC. Velasco & Gaona (2011) — mushroom panel.

Standards: ASTM D790-17 · ISO 11925-2:2020 · ASTM D570-98(2018).